

SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

**UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II 2022/2023**

SUBJECT CODE : SECP2753
SUBJECT NAME : DATA MINING
SECTION : 01
TIME : 2 HOURS 30 MINUTES
DATE/DAY :
VENUES :

INSTRUCTIONS:

1. Please answer **ALL** the question in the answer sheet form.
2. Fill in your particular in the answer sheet.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **FIVE (5)** printed pages including this page.

QUESTION 1

(25 MARKS)

- a) Data pre-processing and conditioning is one of the key factors that determine whether a data mining project will be a success. For each of the following topics, describe the affect this issue can have on our data mining session and what techniques can we use to counter this problem.
- i) Noisy data
 - ii) Missing data
 - iii) Data normalization and scaling
 - iv) Data type conversion
 - v) Attribute and instance selection

[15 Marks]

- b) Data pre-processing is a data mining technique that involves transforming raw data into an understandable format. There are a few steps or processes involving in data pre-processing. Based on the sample of input shown in Figure 1, discuss the appropriate data pre-processing steps involved in order to produce the final output, as shown in Figure 2.

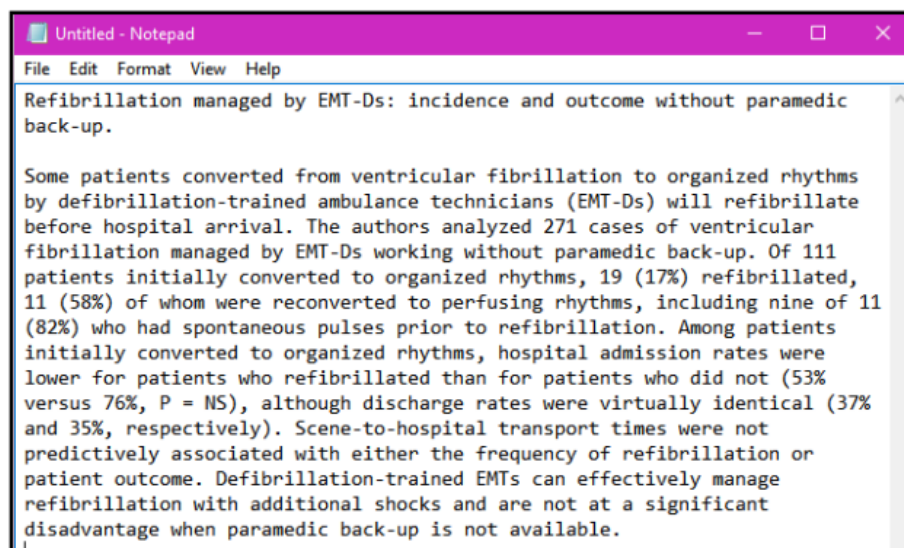
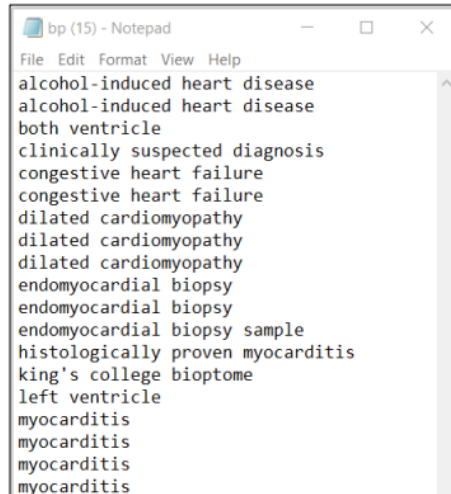


Figure 1: Sample of raw data (input)



```
bp (15) - Notepad
File Edit Format View Help
alcohol-induced heart disease
alcohol-induced heart disease
both ventricle
clinically suspected diagnosis
congestive heart failure
congestive heart failure
dilated cardiomyopathy
dilated cardiomyopathy
dilated cardiomyopathy
endomyocardial biopsy
endomyocardial biopsy
endomyocardial biopsy sample
histologically proven myocarditis
king's college biptome
left ventricle
myocarditis
myocarditis
myocarditis
myocarditis
```

Figure 2: Sample of output

[10 Marks]

QUESTION 2 **(25 MARKS)**

- a) What is classification?[3 Marks]
- b) List **FIVE (5)** data mining algorithms that can be used for classification purpose.[5 Marks]
- c) Give **ONE (1)** example of cases where the data analysis task is classification.[3 Marks]
- d) Based on your answer in (c), explain how does classification work.[6 Marks]
- e) Differentiate between supervised learning and unsupervised learning.[8 Marks]

QUESTION 3**(25 MARKS)**

- a) Based on Figure 3 below, are these two clusters well separated? Justify your answer.

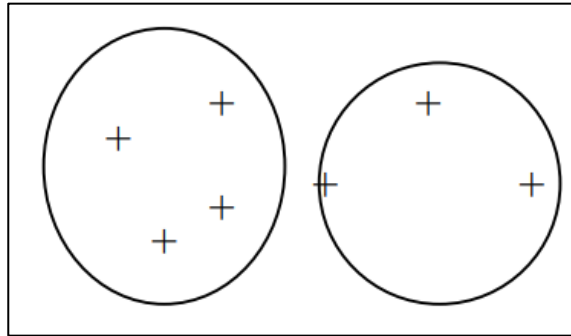


Figure 3: Clustering

[3 Marks]

- b) Briefly describe (give an example) each of the following approaches to clustering:
- i) partitioning method
 - ii) hierarchical method

[8 Marks]

- c) Suppose that the data mining task is to cluster points (with (x, y) representing location) into three clusters, where the points are given in the Table 1;

Table 1

SUBSET	x_i	x_j
A	2	10
B	2	5
C	8	4
D	5	8
E	7	5
F	6	4
G	1	2
H	4	9

The distance function is Euclidean distance. Suppose initially we assign A, D, and G as the center of each cluster, respectively. Use the k-means algorithm to show;

- i) the three cluster centers after the first round of execution.

[8 Marks]

- ii) the final three clusters.

[6 Marks]

QUESTION 4**(25 MARKS)**

- a) Define support and confidence in Association rule mining. How are Association rules mined from large databases?
[10 Marks]
- b) List **THREE (3)** example of applications of Association rule mining?
[3 Marks]
- c) Table 2 shows the transactions for generating strong rules from frequent itemset.

Table 2

<i>TID</i>	<i>items_bought</i>
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y }
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I ,E}

- i) Find all frequent itemsets using Apriori and FP-growth algorithms, respectively.
[10 Marks]
- ii) Based on your answer in (c) (i), compare the efficiency of these two algorithms.
[2 Marks]